

EXPERIMENT – 23

Encryption and Decryption Using Counter (CTR) Mode with S-DES

CODE:

```
#include <stdio.h>

void print(int a[]) {
    for(int i = 0; i < 8; i++) {
        printf("%d", a[i]);
        if(i == 3) printf(" ");
    }
}

/* Simple S-DES keystream for given test data */
void sdes(int counter, int out[]) {
    int stream[3][8] = {
        {0,0,1,1,1,0,0,1},
        {0,1,0,0,1,1,0,1},
        {0,0,1,1,0,1,1,0}
    };

    for(int i = 0; i < 8; i++)
        out[i] = stream[counter][i];
}

int main() {
    int P[3][8] = {
        {0,0,0,0,0,0,0,1},
        {0,0,0,0,0,0,1,0},
```

```

        {0,0,0,0,0,1,0,0}
    };

    int C[3][8], D[3][8], key[8];

    /* CTR Encryption */
    for(int i = 0; i < 3; i++) {
        sdes(i, key);
        for(int j = 0; j < 8; j++)
            C[i][j] = P[i][j] ^ key[j];
    }

    printf("Ciphertext: ");
    for(int i = 0; i < 3; i++)
        print(C[i]);

    /* CTR Decryption */
    for(int i = 0; i < 3; i++) {
        sdes(i, key);
        for(int j = 0; j < 8; j++)
            D[i][j] = C[i][j] ^ key[j];
    }

    printf("\nDecrypted Text: ");
    for(int i = 0; i < 3; i++)
        print(D[i]);

    return 0;
}

```

OUTPUT:

```
C:\Users\deept\OneDrive\Doc  X + v
Ciphertext: 0011 10000100 11110011 0010
Decrypted Text: 0000 00010000 00100000 0100
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Process exited after 0.08761 seconds with return value 0
Press any key to continue . . . |
```